

ing assembly of QA/QC activities due to which component failure may occur

• Soldering problems

To cover the fabrication/assembly process and to measure the quality of manufacturing, there are some testing measures that need to be followed, a few of which are mentioned below:

- PCB analysis test
- Zero-value analysis
- Soldering evaluation and corrective action

A few examples of the type of quality non-conformance and damages that may occur in the manufacturing process and in the field are shown in Figs 2 through 6.

There are many dependencies in the manufacturing of consumer electronics products. Examples of dependencies are given below:

- Dependencies with system-on-chip manufacturer and vendor
- Dependencies with raw materials like small electronic and electrical components
- Dependencies with packaging vendors
- Dependencies with product body manufacturing vendors
- Dependencies with PCB vendors
- Dependencies with PCB layout vendors
- Dependencies with logistics vendors

Despite these dependencies, it is a great challenge for manufacturing industries to sync with various vendors and maintain manufacturing process quality. Following are some possible impacts if challenges are not met appropriately:



Fig. 3: Shorts between components and pins

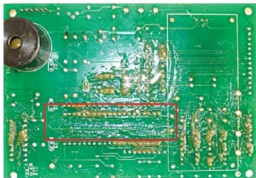


Fig. 4: Dust on PCB after soldering



Fig. 5: Ant on PCB surface

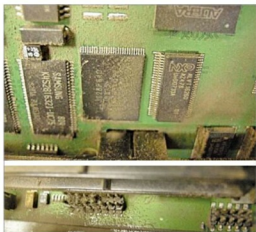


Fig. 6: Dust and debris in memory ICs

- Increase in field call returns
- Increase in rework
- Increase in warranty cost
- Increase in scrap
- Increase in low performance in overall productivity

To address the above points, non-conformance management systems should ask the following questions during the manufacturing process:

- How long does it take to find, analyse and resolve manufacturing issues?
- Are you sure you have really solved the issue?
- Is the issue recurring?
- How do you implement and enforce preventive measures?
- Do you really have the data required to identify non-conformances?

Non-conformance management systems teams usually work on the following mindsets:

- To identify any deviations in processes
- To identify quality assurance test results on product, packaging and product body
- To initiate appropriate actions to overcome the above two points
- To initiate field call returns analysis and take appropriate actions depending on analysis reports

Cost due to non-conformance in raw materials.

To make consumer electronics products, we need many raw materials in the form of electronic, electrical, mechanical and packaging materials. These call for the involvement of multiple vendors to supply those raw materials. Hence, manufacturers need to evaluate the caliber of new materials that various vendors are offering. Examples given below show the problems that may occur due to non-compliance of raw materials:

- Failure of electronic/electrical components
- Damage of products during transportation due to inferior packaging quality